PH PHEWPHONICS · MAD FRAMEWORK

MAD RUN — OCTOBER 2026 · AA · DA · SSA

Autodesk Vault Cloud Migration Findings

Multi-Agent Debate covering Inventor, Plant 3D, AutoCAD, PDF, STEP/IGES, IFC, and Excel BOM. Thirteen critical failure modes identified across all integration surfaces. Verdict synthesised from full AA and DA debate cycles.

● CONDITIONAL GO

VAULT PROFESSIONAL | ON-PREM → CLOUD | AA · DA · SSA | PLANT 3D HIGHEST RISK | OCTOBER 2026

SECTION 1

SSA Verdict

Final synthesised output from the Multi-Agent Debate cycle

SSA FINAL VERDICT

Conditional Go — Six Conditions Must Be Met

The migration is technically viable and Autodesk-supported. The Vault stack on a cloud VM is the same software as on physical hardware. All file formats are preserved through the SQL + file store migration. However, 13 critical failure modes exist that standard IT testing will not catch — nine specific to Plant 3D and Inventor workflows. Until six named conditions are satisfied, go-live carries unacceptable operational risk.

13	8	6	0
CRITICAL MODES	SIGNIFICANT	GO CONDITIONS	IT TESTS CATCH

- ✓ **Technically possible** — Vault Professional runs legitimately on cloud VMs (AWS + Azure), Autodesk-documented and supported
- ✓ **All file formats preserved** — SQL + file store migration maintains revision history, lifecycle states, BOM, and properties
- ✓ **Plant 3D constraint unchanged** — check-in direction constraint is a Plant 3D app behaviour; cloud doesn't worsen it technically
- ! **IT test cases are insufficient** — both AA and DA agree standard IT migration testing will miss the highest-risk scenarios

SECTION 2

13 Critical Failure Modes

All identified by the DA agent. None are caught by standard IT migration testing.

#	FAILURE MODE	APP	RATING	IT?
1.1	Check-in direction reversal triggered by path/UX change	PLANT 3D	CRITICAL	×
1.2	Project.xml hostname reference invalidation	PLANT 3D	CRITICAL	×
1.3	Lock file race condition under cloud latency	PLANT 3D	CRITICAL	×
2.1	300-part assembly open time at concurrent latency (21–42s added)	INVENTOR	CRITICAL/SIG	×
2.2	iLogic Vault property reads returning null	INVENTOR	CRITICAL	×
2.3	Vault add-in version mismatch — day-one outage for all users	ALL APPS	CRITICAL	~
5.2	SQL item number change breaking BOM history permanently	SQL	CRITICAL IF OCCURS	×
6.1	SQL relationship graph corruption in staged migration	SQL	CRITICAL	×
6.2	SQL config still pointing to old server hostname	SQL	CRITICAL	×
6.3	Missing binary files for historical revisions	SQL	CRITICAL	×
7.1	Cumulative latency under concurrent peak load (60ms+)	INVENTOR	CRITICAL ≥60MS	×
8.1	Rollback data loss — two weeks of work, untested procedure	ALL	CRITICAL	×
9.1	Test cases written without Plant 3D operator input	PLANT 3D	CRITICAL (META)	×

Key: × = IT test will not catch this | ~ = IT may catch with per-machine testing only

SECTION 3

6 Go / No-Go Conditions

All six must be met before go-live. Binary — each is either satisfied or it is not.

1 Latency Measured & Within Threshold

Actual round-trip from every workstation through corporate proxy to cloud Vault. ≤40ms → Go. 40–70ms → AVFS cache required. >70ms → escalate to leadership. Largest 3 assemblies timed with 10 concurrent users.

2 Plant 3D Test Cases Signed Off by Operator

A Plant 3D daily user — not IT — must review and sign off the test case list. Must cover: check-in direction test, project.xml paths, lock file concurrency, isogen output, Data Manager connection, pipe spec catalog.

3 Version Matrix Confirmed Per Workstation

Vault client add-in version documented for every machine. Cross-referenced against Autodesk's official compatibility matrix for the cloud server version. Written confirmation from cloud provider or VAR.

4 Project.xml Path Remediation Documented & Tested

Confirm whether working folder path changes. If yes: audit all Plant 3D project.xml files, document every edit required, execute remediation in test environment, confirm every project opens successfully.

5 Rollback Procedure Executed in Test Environment

Run the rollback — not just document it. Confirm all files recoverable. Define maximum acceptable data loss window. Get engineering management sign-off in writing before migration proceeds.

6 All Checked-Out Files Cleared

Audit every checked-out file across all users. Define resolution procedure for each engineer. Hard technical prerequisite — migration does not begin until all files are checked in.

Highest single risk: Condition 2 — Plant 3D test cases not authored by a Plant 3D operator. This is the meta-risk that causes all other risks to go undetected. If only one condition can be prioritised, it is this one.

SECTION 4

Strangler Fig Cutover Plan

Phase by application — lowest risk first, Plant 3D last. On-premise set to read-only at each phase.

WEEK 1–2	WEEK 3–4	WEEK 5–6	DAY 60+
AutoCAD	**Inventor**	**Plant 3D**	**Decommission**
Lowest risk. Fastest to verify xref resolution. Any path issues surface here at lowest cost.	Latency monitoring active. Assembly open times tracked. iLogic rules tested. AVFS in place if required.	All project.xml remediation complete. Plant 3D operator on-site day one. Lock file behaviour monitored.	60 days after Plant 3D cutover with no open rollback issues.

Rollback Triggers

Pause and escalate immediately if any of these occur

- **Plant 3D check-in fails for any user post-go-live** — Pause Plant 3D cutover; investigate path/hostname before proceeding
- **Assembly open time >90 seconds for any user** — Activate AVFS evaluation immediately; do not proceed with Inventor cutover until resolved
- **Any Vault add-in reports "cannot connect to server"** — Full stop; verify SQL hostname config and version matrix
- **Any SQL consistency check error post-migration** — Full stop; resolve before any user check-ins
- **Historical revision opens with "file not found"** — Log immediately; do not proceed with decommission until all instances resolved
- **Two concurrent Plant 3D check-ins to same file both succeed** — Treat as critical data integrity event; audit both versions before further check-ins

SECTION 5

8 Priority Test Scenarios

Run in this order. Run before go-live — not during. Document every deviation from on-premise behaviour.

00 PRE-TEST

Version Matrix — Before Anything Else

Confirm cloud Vault server version matches the Vault client add-in on your test machine. If they don't match — stop and escalate. A mismatch invalidates all subsequent test results.

01 PLANT 3D

Wrong-Direction Test

Open a Plant 3D drawing from Vault Explorer (not from within Plant 3D). Edit it. Attempt to check in. Document exactly what happens — failure, success, or silent corruption.

02 PLANT 3D

Concurrent Editing Race

Two users open the same P&ID simultaneously. Both edit and attempt to check in. Does the second user see the first user's lock file? What happens to the overwritten version?

03 INVENTOR

Large Assembly Timing

Open the three largest assemblies with 5–10 people simultaneously doing other Vault operations. Record time from click to fully loaded. Measure latency from each workstation separately.

04 INVENTOR

iLogic Rule Execution

Find drawings with iLogic rules reading Vault properties. Run each rule on the cloud system. Verify returned values match what Vault actually shows in the web client.

05 PLANT 3D

Isometric Generation Path

Run isogen on a known P&ID. Confirm output appears in Vault under the correct revision. Compare content line-by-line against a known-good reference from the on-premise system.

06 ALL

Historical Revision Open

Find drawings with 5+ revisions. Open each old revision from Vault history. Confirm all open — not just the latest. This is the binary file integrity test for the migration.

07 AUTOCAD

Sheet Set Batch Plot

Locate all .dst sheet set files. Run batch plot on each. Confirm all sheets plot and output is correct. Silent path failures only surface during batch plot operations.